CHRONIC PAIN & FUNCTIONAL NEUROLOGICAL DISORDER

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Slides adapted from: Irvind Sihota

LEARNING OUTCOMES

- Demonstrate an understanding of the transmission system for nociception.
- Describe the relationship between pain and tissue damage
- Explain the basic concept of pain as a multisystem body response to danger or damage
- Describe the basic responses to the experience of pain and factors that lead to the development of chronic pain.

WHAT IS CHRONIC PAIN?

- **Chronic Pain** - Pain that persists or recurs for >3 months. It outlasts the normal healing time for tissues.
  - Chronic pain is not a "symptom" of a broken part; it is a malfunction of the nervous system itself.
- **Nociplastic Pain** – Pain that arises from altered nociception despite no clear evidence of actual or threatened tissue damage.
- **Functional Neurological Disorder (FND)-** A condition where the brain sends/receives signals incorrectly (e.g., functional tremors, paralysis, or seizures) despite the "hardware" (nerves/muscles) being intact.

- Both Chronic Pain and FND are "Software" problems. The brain's predictive processing is "glitched," creating a real physical experience (pain or movement loss) without a structural "break."
- In both cases, the nervous system has become hyper-vigilant. It is reacting to "threats" that aren't actually there, similar to a car alarm that goes off when a leaf touches the windshield.
- While both are 'software' issues, they are classified by their primary output: **Nociplastic Pain** is a disorder of **sensation** (how the brain feels), whereas **FND** is a disorder of **function and control** (how the brain moves or regulates the body). In modern coding (ICD-11), Chronic Pain is now recognized as a primary disease entity, while FND remains classified under neurological/psychobiological processing disorders.

- A patient can have **Chronic Low Back Pain** (the location) caused by **Nociplastic** mechanisms (the "how"), presenting with **FND** symptoms (the "expression").

HOW DOES IT HAPPEN?



Pillar 1:Peripheral Sensitization

- Happens in the tissues (nerve endings)
- Local inflammation creates a "chemical soup" (Prostaglandins, Histamine, Cytokines) around nociceptors.
- Things that shouldn't hurt (like the touch of clothing) now trigger a "danger" signal. The alarm system is now "twitchy."

Pillar 2: Central Sensitization

- Happens in the Spinal Cord (Dorsal Horn)
- Persistent Bottom-Up signals cause the spinal cord to increase its spontaneous firing. This is called **Wind-Up**.
- The spinal cord becomes a loudspeaker. Even if the injury heals, the "volume knob" is stuck at 10, sending constant alerts to the brain.

Pillar 3: Neuroplasticity

- Happens in brain
- The brain physically rewires itself to become better at feeling pain. The areas responsible for emotion, fear, and memory (Amygdala/Hippocampus) link up with the pain centers.
- Pain is no longer about the tissue; it’s about **Threat Evaluation.** The brain predicts pain before movement even happens.

NORMAL HEALING

Step 1: The Initial Injury (Peripheral) - Inflammation occurs to protect the site, then subsides as tissues repair.

Step 2: The spinal cord amplifies these signals (BRIEFLY) - The spinal cord increases sensitivity briefly to ensure you don't use the injured limb (Protective guarding).

Step 3: The Reset - Once the tissue heals, Bottom-Up signals stop. The brain recognizes the threat is over and sends **Descending Inhibition** (the brakes) to reset the system.



ALTERED HEALING

The system fails to "down-regulate" even after the tissue has structurally healed:

Step 1: The Initial Injury - Injury or stress triggers a Bottom-Up alarm. Nociceptors become "twitchy" (Pillar 1).

Step 2: The spinal Cord amplifies these signals - The spinal cord is bombarded by these "twitchy" signals. It decides to "help" by amplifying every signal it receives. This is **Wind-Up** (Pillar 2).

Step 3: Neuroplasticity - The brain sees the constant spinal activity and decides this is a "Survival Crisis." It strengthens the neural pathways for pain so it can detect the "threat" faster (Pillar 3).

Step 4: Top Down Failure - Because the brain is now in a "protective state, it stops sending descending brakes (opioids/serotonin) and starts sending facilitation (gas like neurotransmitters and facilitatory pathways) to the spinal cord, making Pillars 1 and 2 even worse.

HOW THEY DIFFER

Stage	Normal Healing (Acute)	Chronic/Nociplastic Path
1. Initial Injury	Signal stops once tissue repairs.	Signal persists or the nerve stays "twitchy."
2. Spinal Cord	Returns to normal baseline.	**Wind-Up:** Neurons stay hyper-excitable long-term.
3. Neuroplasticity	Brain maps remain stable.	**Smudging:** Cortical maps for the body part blur/expand.
4. Top-Down	Brain "vetoes" minor aches.	**Disinhibition:** The "brakes" fail; the brain amplifies noise.

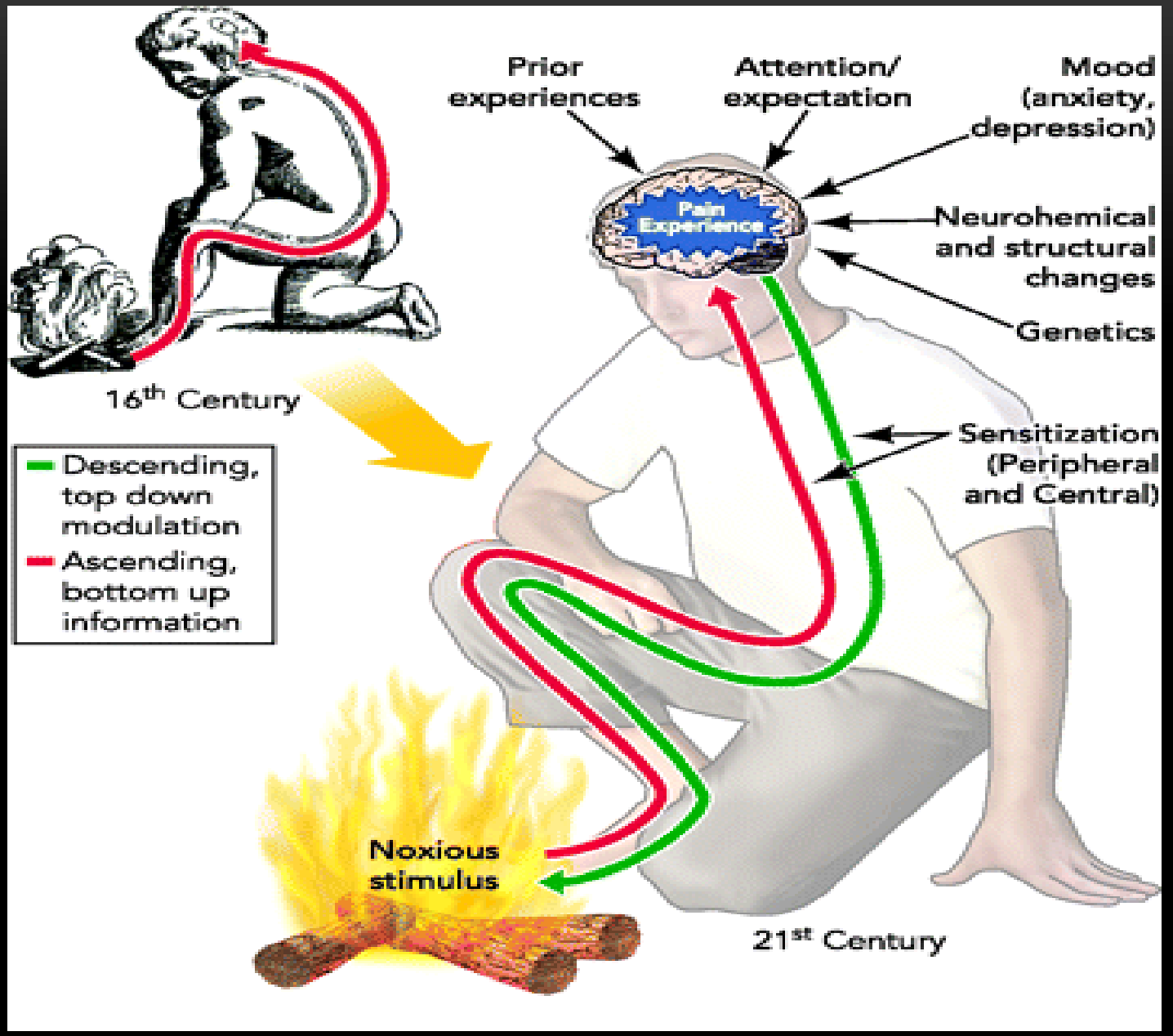
HOW MIGHT PATIENTS PRESENT?

- 1. Secondary Hyperalgesia:
  - The patient reports pain in a much larger area than the original injury.
  - Pain spreads to areas that weren't even injured (The "Volume" is so loud it leaks into neighboring nerves).
  - If a patient had a right-sided L5 disc issue but now has pain across both hips and up into their mid-back, you might be seeing Secondary Hyperalgesia.

- 2. Cognitive-Emotional Link:
  - The patient becomes anxious or depressed or "catastrophizing" (the feeling that the pain is a disaster that will never end).
  - This isn't "just a reaction" to pain—it's part of the Pillar 3 rewiring. The brain's emotion centers and pain centers are now physically connected.

- 3. Cortical Map Degradation
  - The patient may have weakness, "giving way" of limbs, tremors, or a loss of coordination that doesn't match nerve distribution.
  - The brain becomes so focused on protection that it forgets how to move the area normally, leading to the functional tremors or weakness seen in FND.

Why the biomedical model fails?



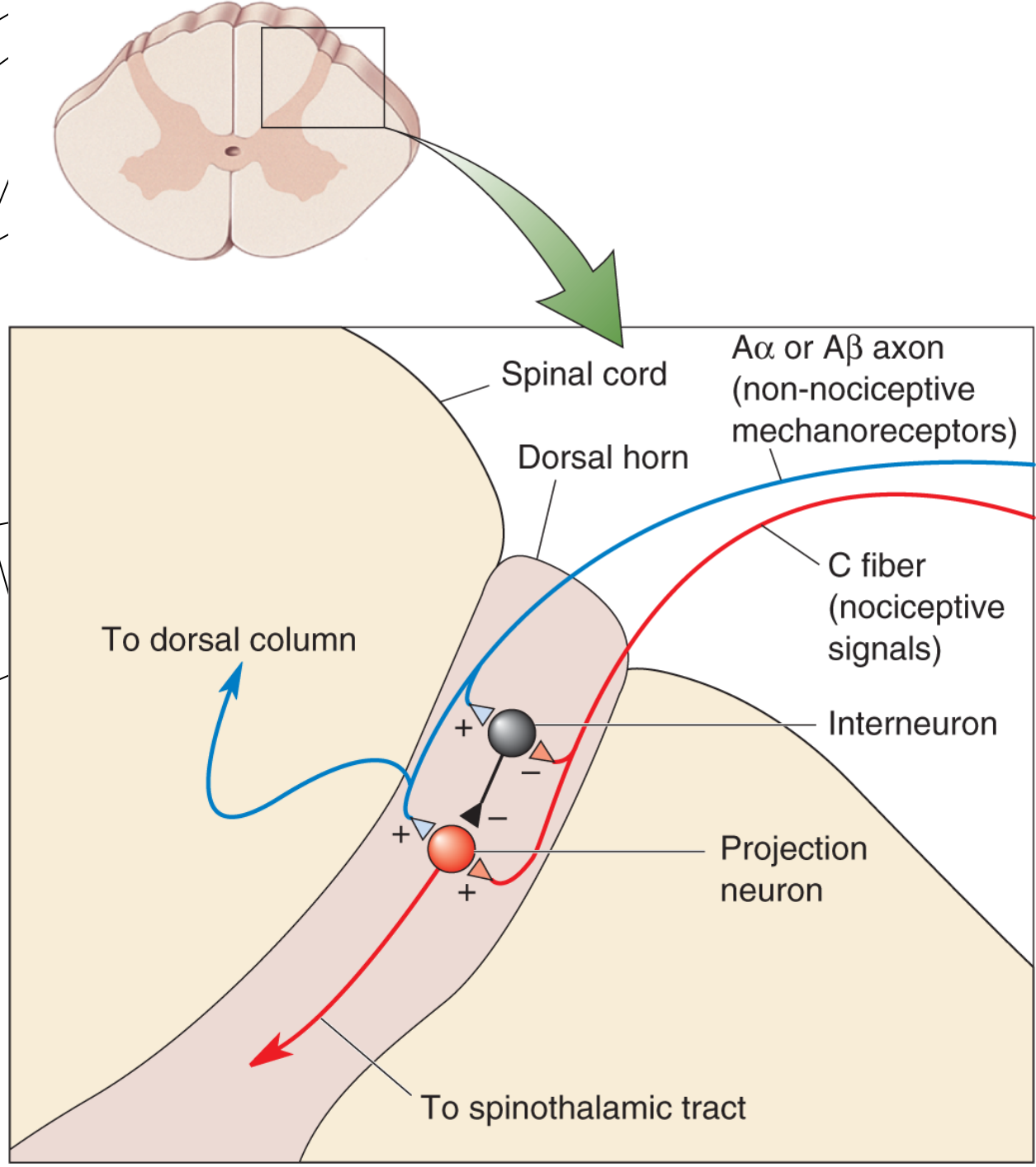
TRANSMISSION SYSTEM FOR NOCICEPTION

&

RELATIONSHIP BETWEEN PAIN AND TISSUE DAMAGE

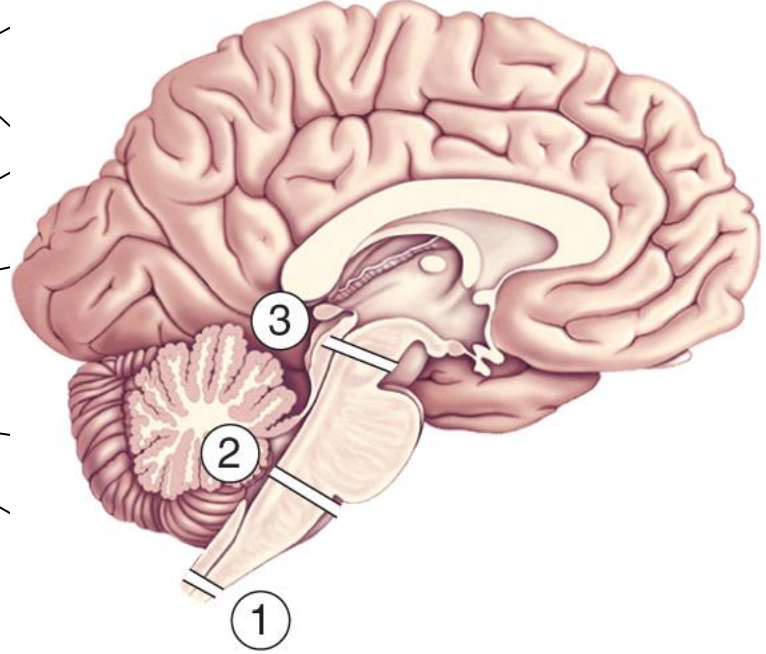
Our Pain Pathway

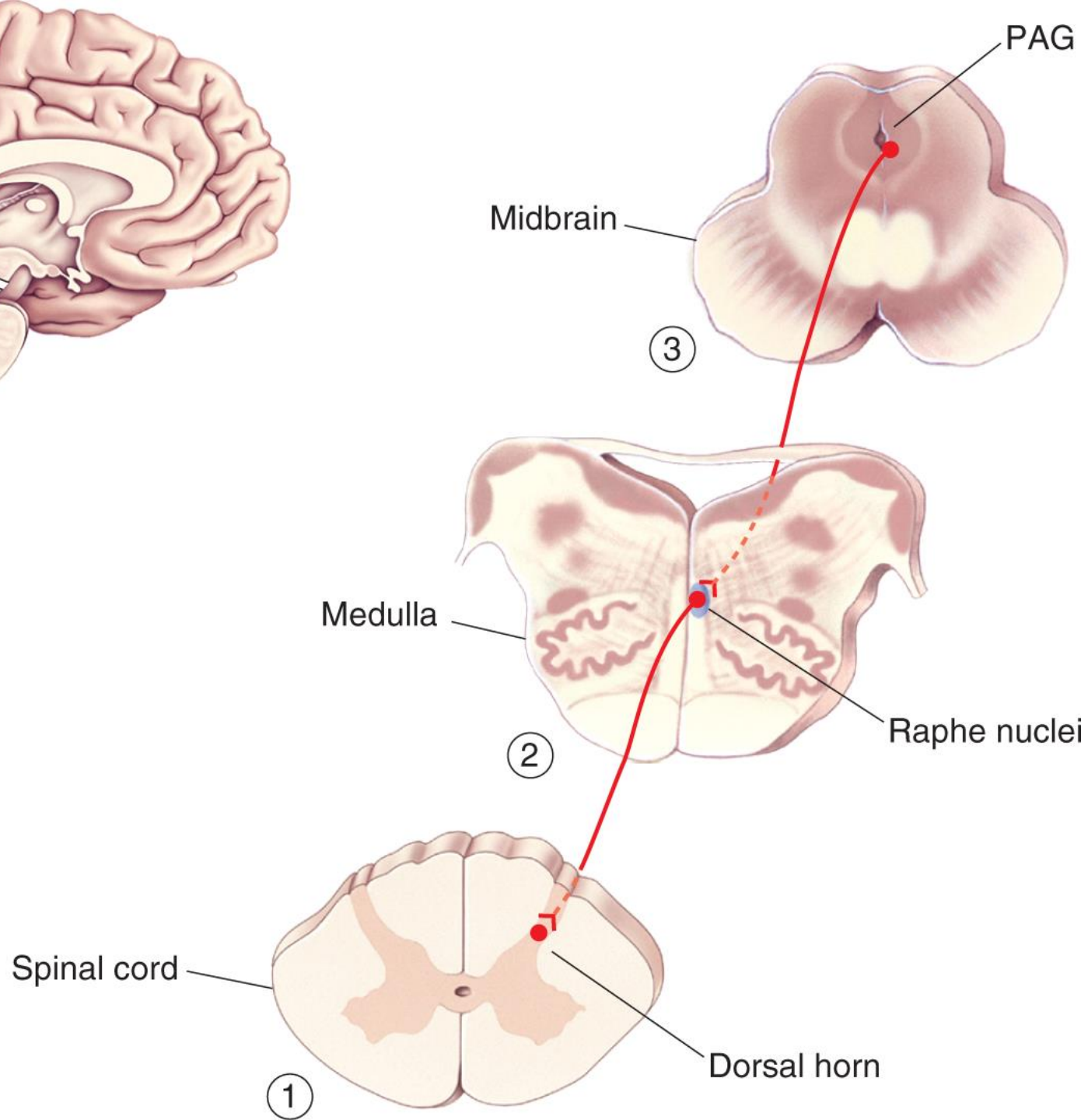
FIGURE 12.33 Melzack's and Wall's gate theory of pain. The relay of nociceptive signals by the projection neuron is gated by the activity of an inhibitory interneuron. Activity in the non-nociceptive mechanoreceptor can suppress, or close the "gate" on, nociceptive signals before they can proceed to the spinothalamic tract. The + signs indicate excitatory synapses and the – signs indicate inhibitory synapses.

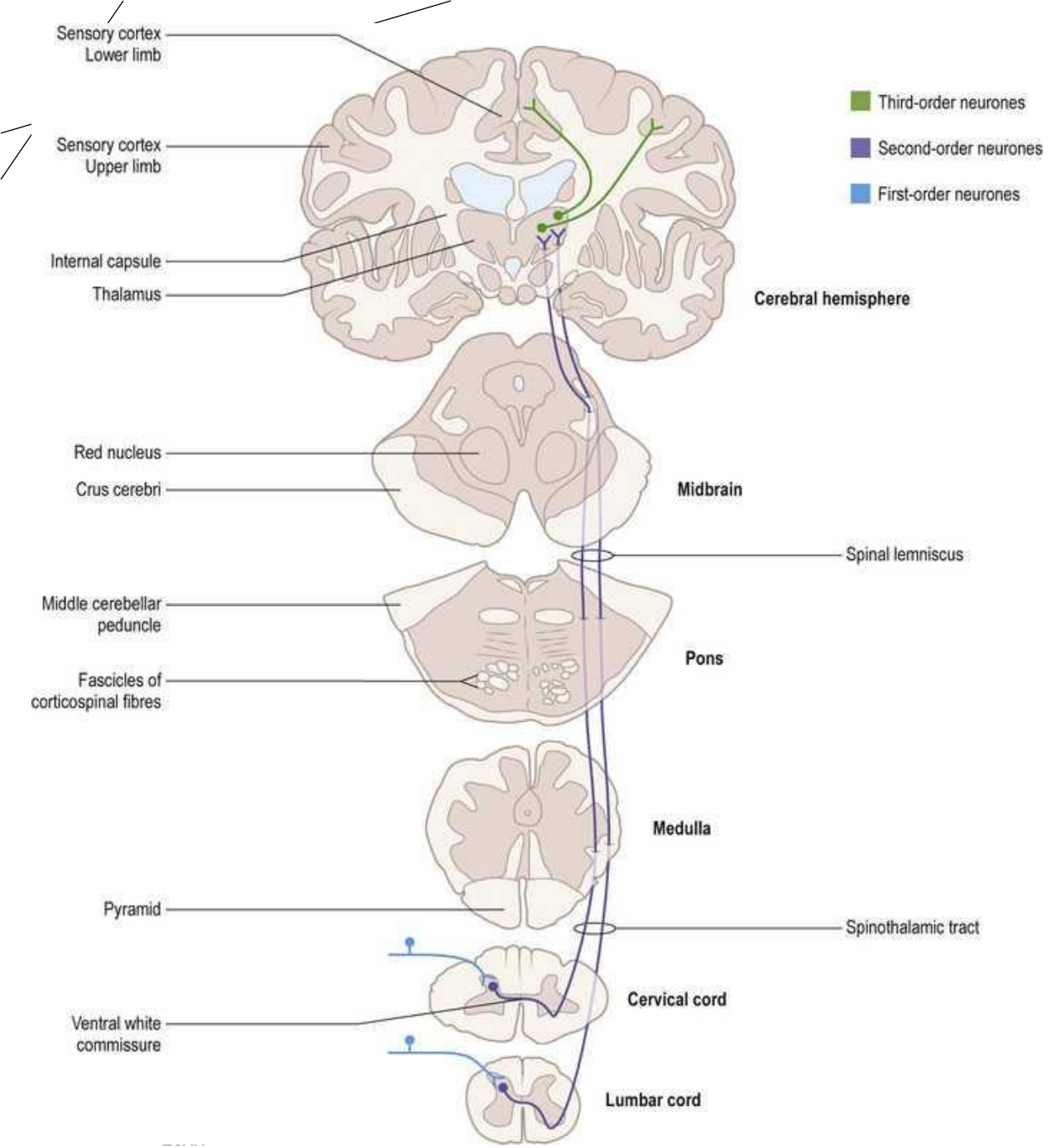


Top Down Control

Descending pain-control pathways. A variety of brain structures, many of which are affected by behavioral state, can influence activity within the periaqueductal gray matter (PAG) of the midbrain. The PAG can influence the raphe nuclei of the medulla, which in turn can modulate the flow of nociceptive information through the dorsal horns of the spinal cord.

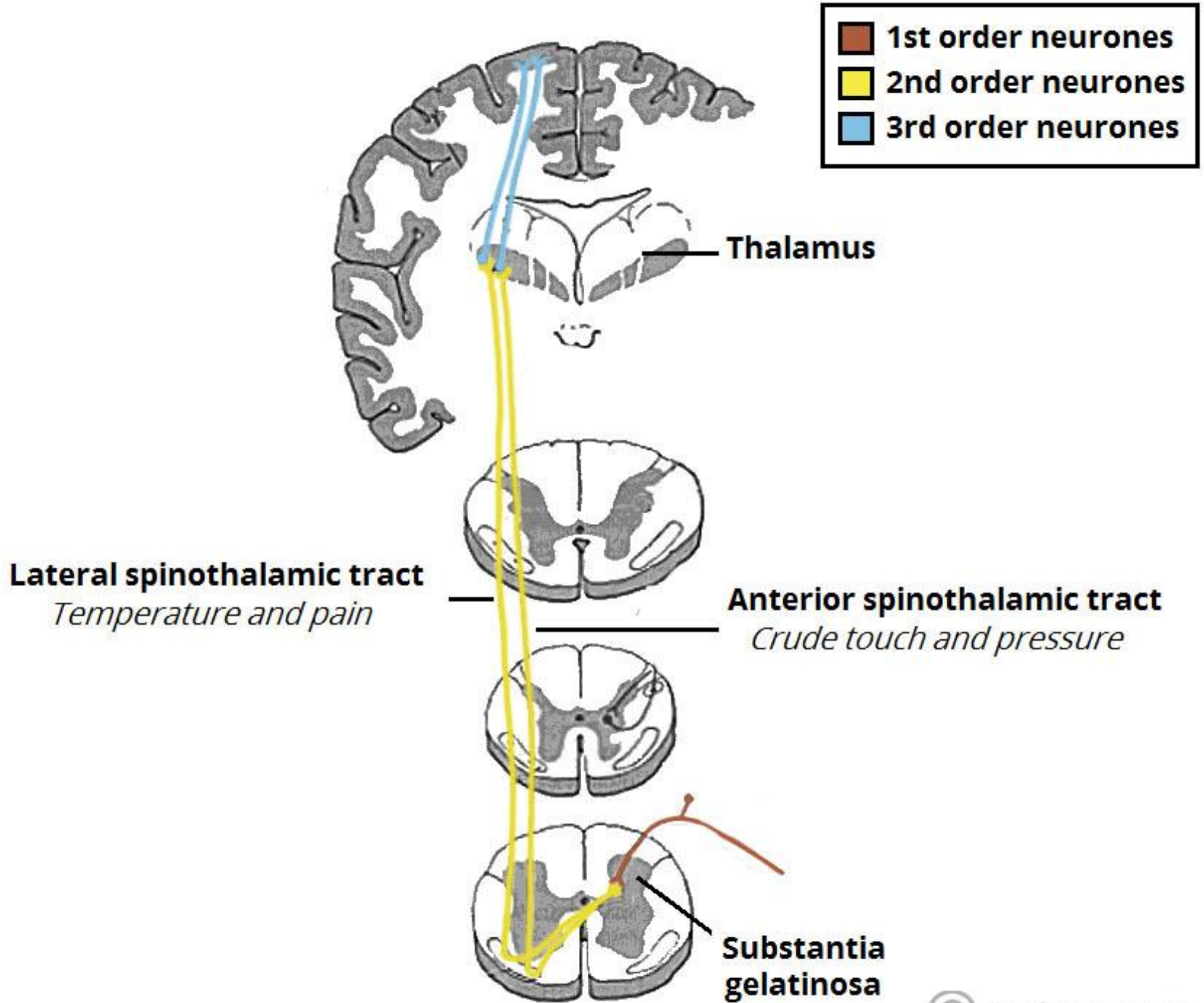






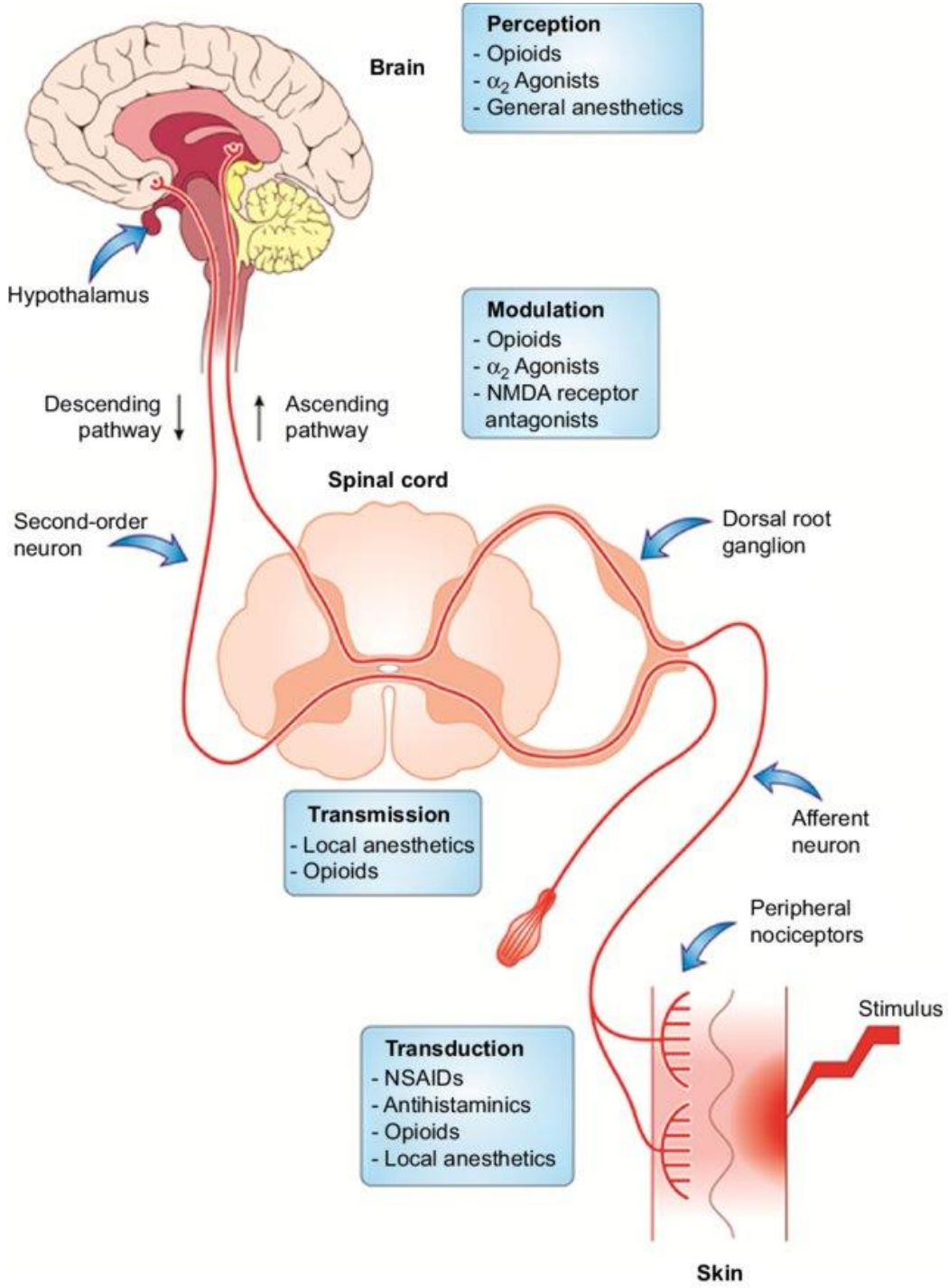
Bottom Up Control

Spinothalamic tract. The central pathways for pain, temperature, touch and pressure are illustrated.

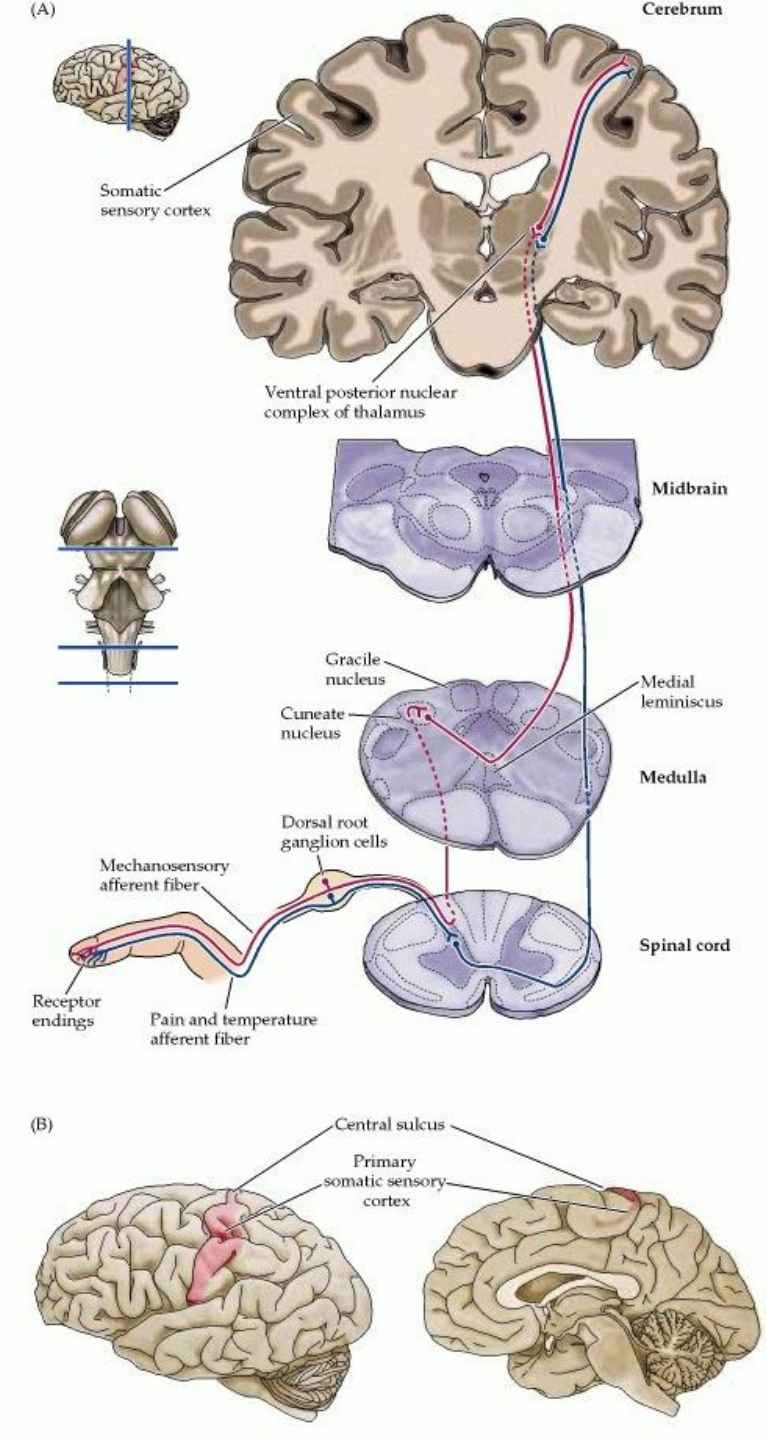


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Level and effect of Medications



Competition between touch (Manual Therapy) and pain



Pain signals can be interrupted in the substantia gelitanosa of the spinal cord, which acts as a "gate".



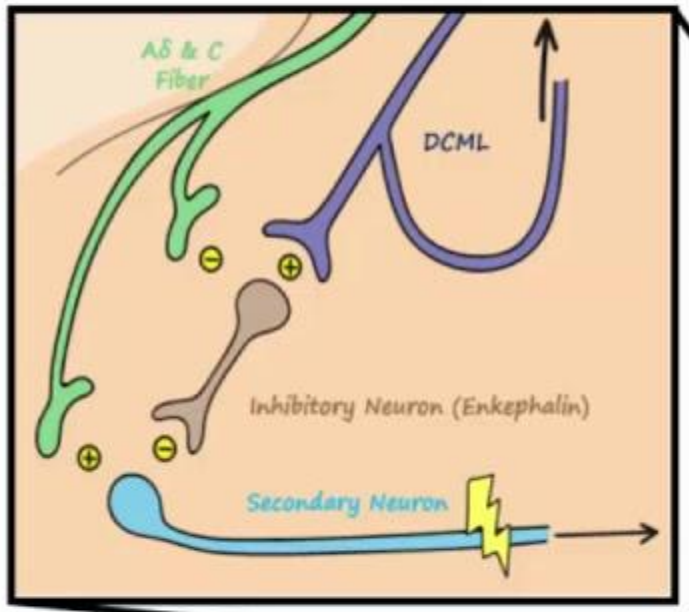
Deep touch

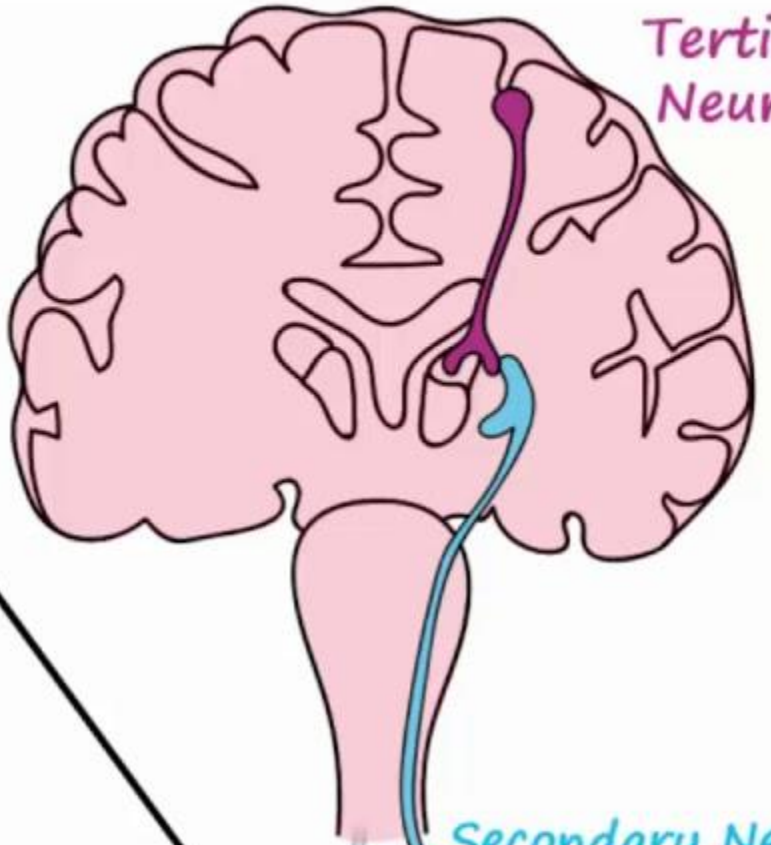
Painful stimulus

Pacinian corpuscle

Aδ and C Fibers

Substantia Gelitanosa



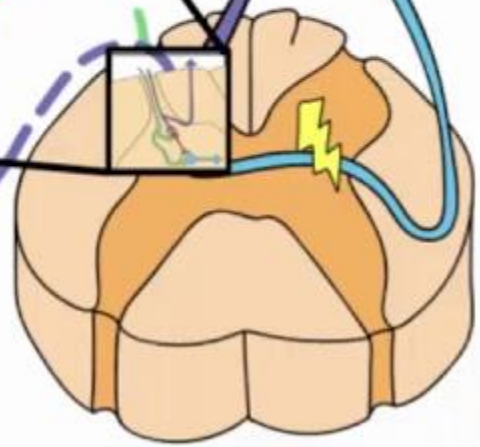


Tertiary Neuron

Secondary Neuron

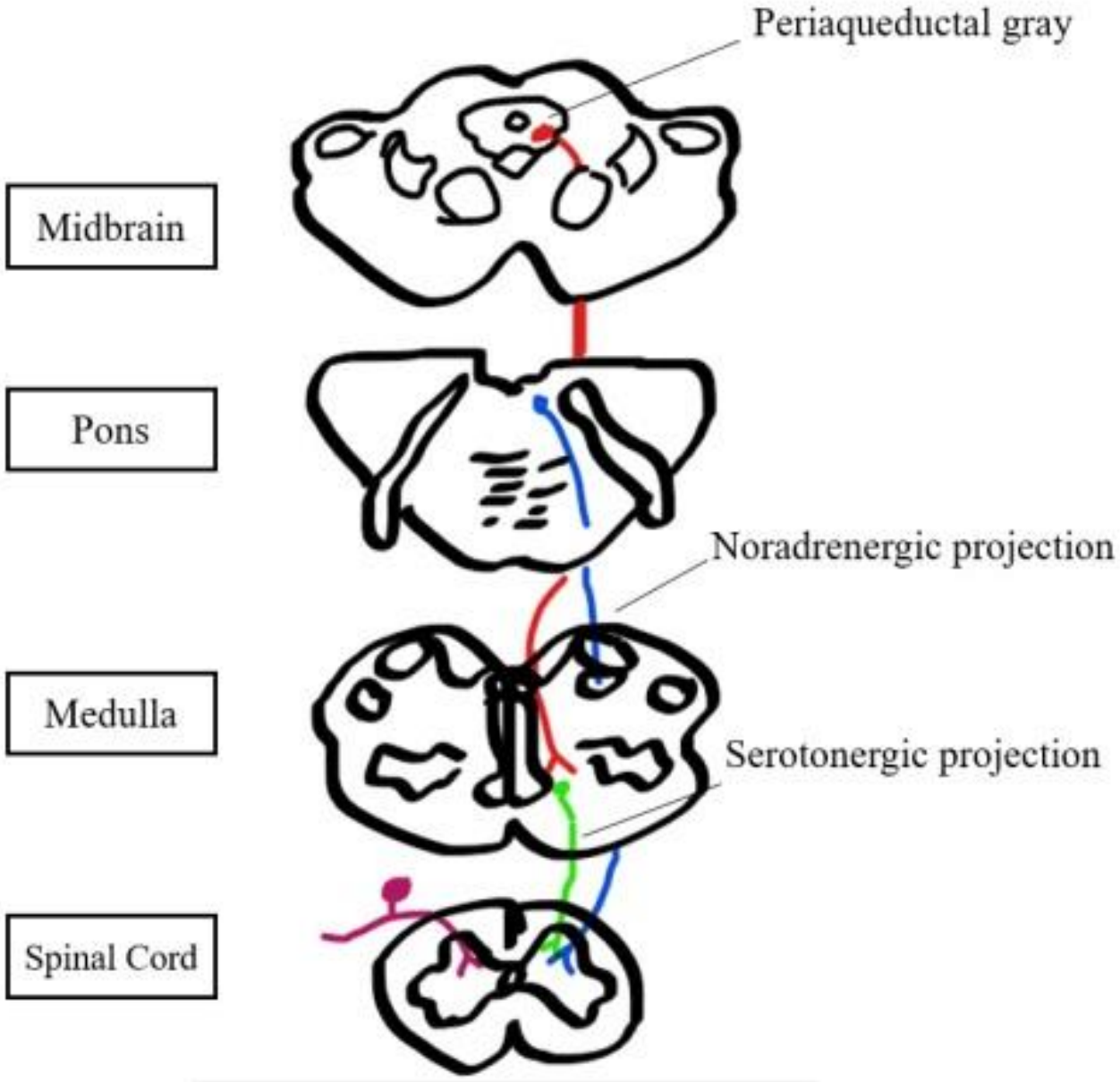
DCML

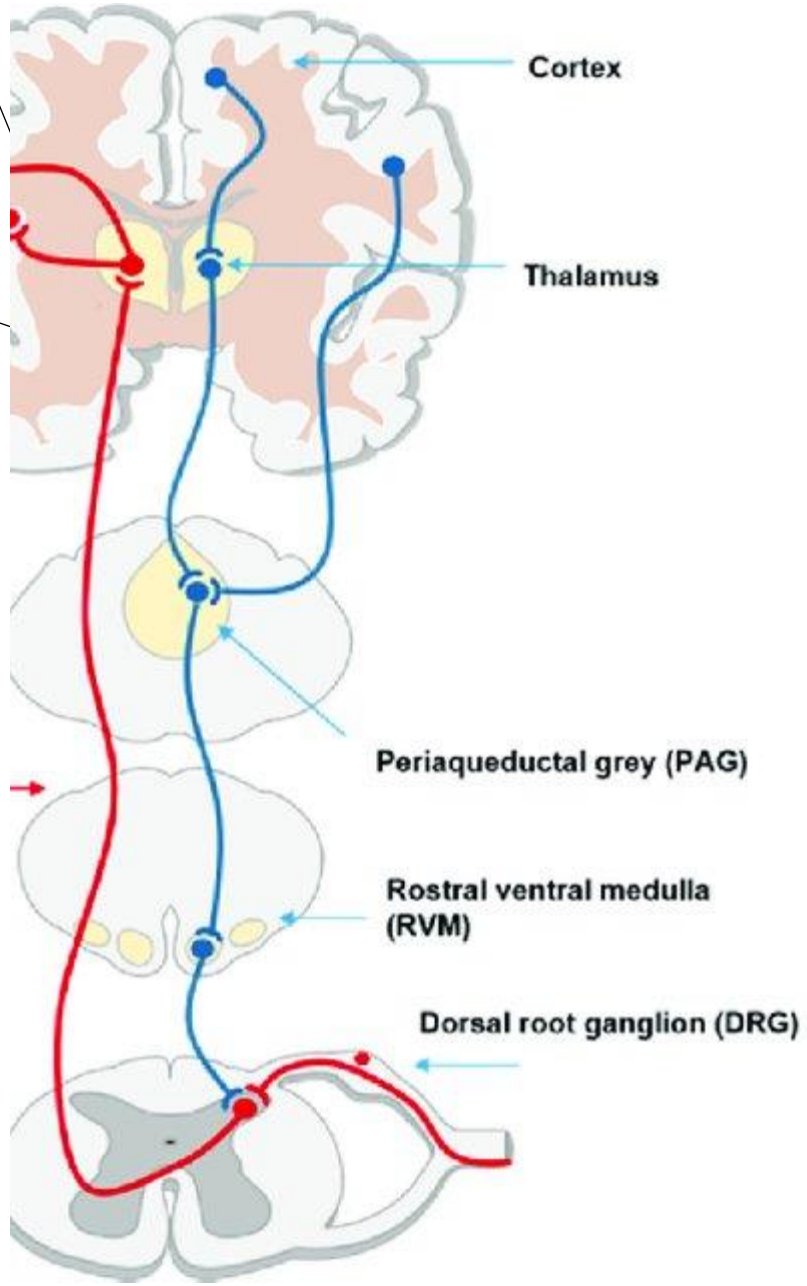
Primary Neuron



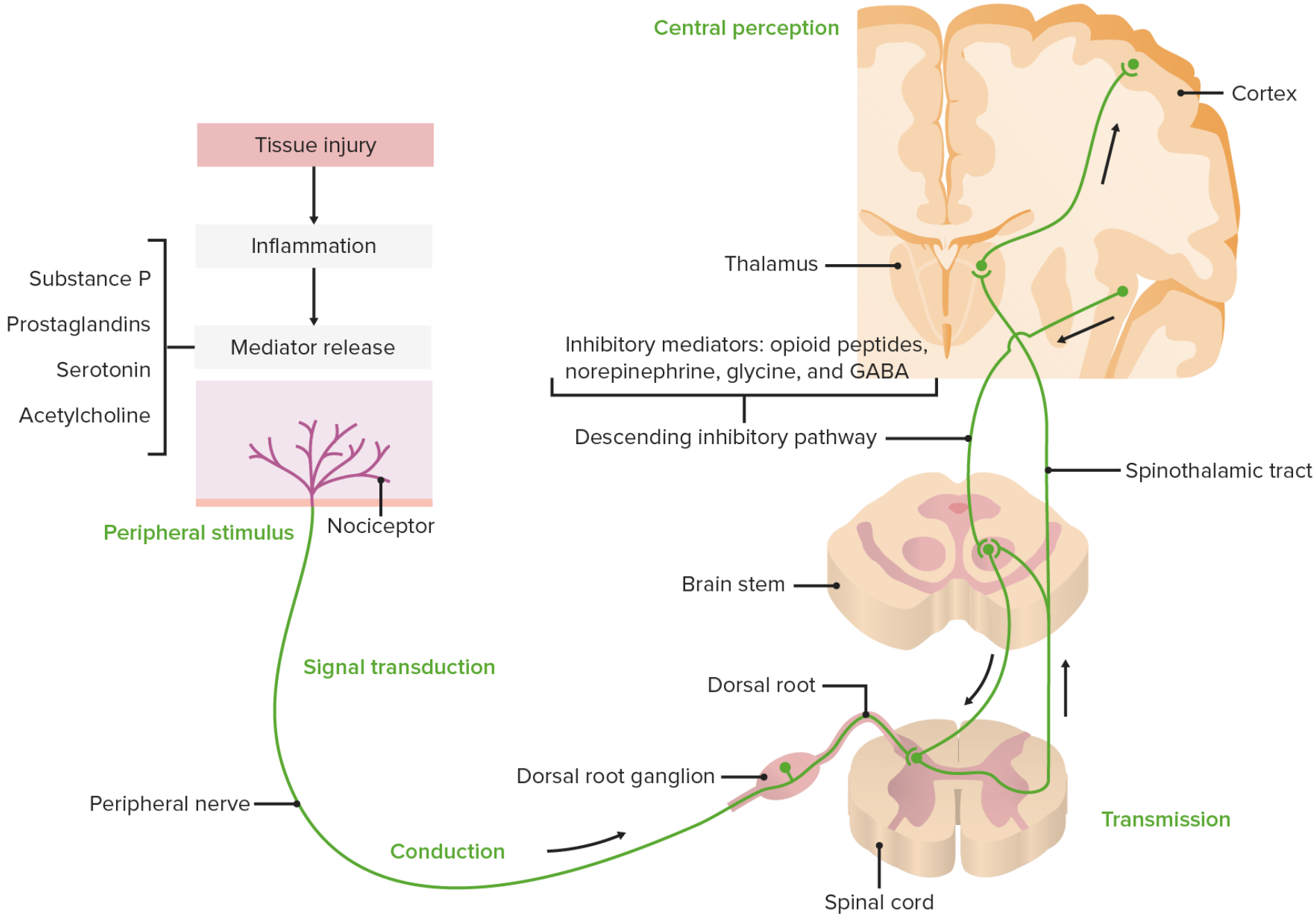
Spinal Cord

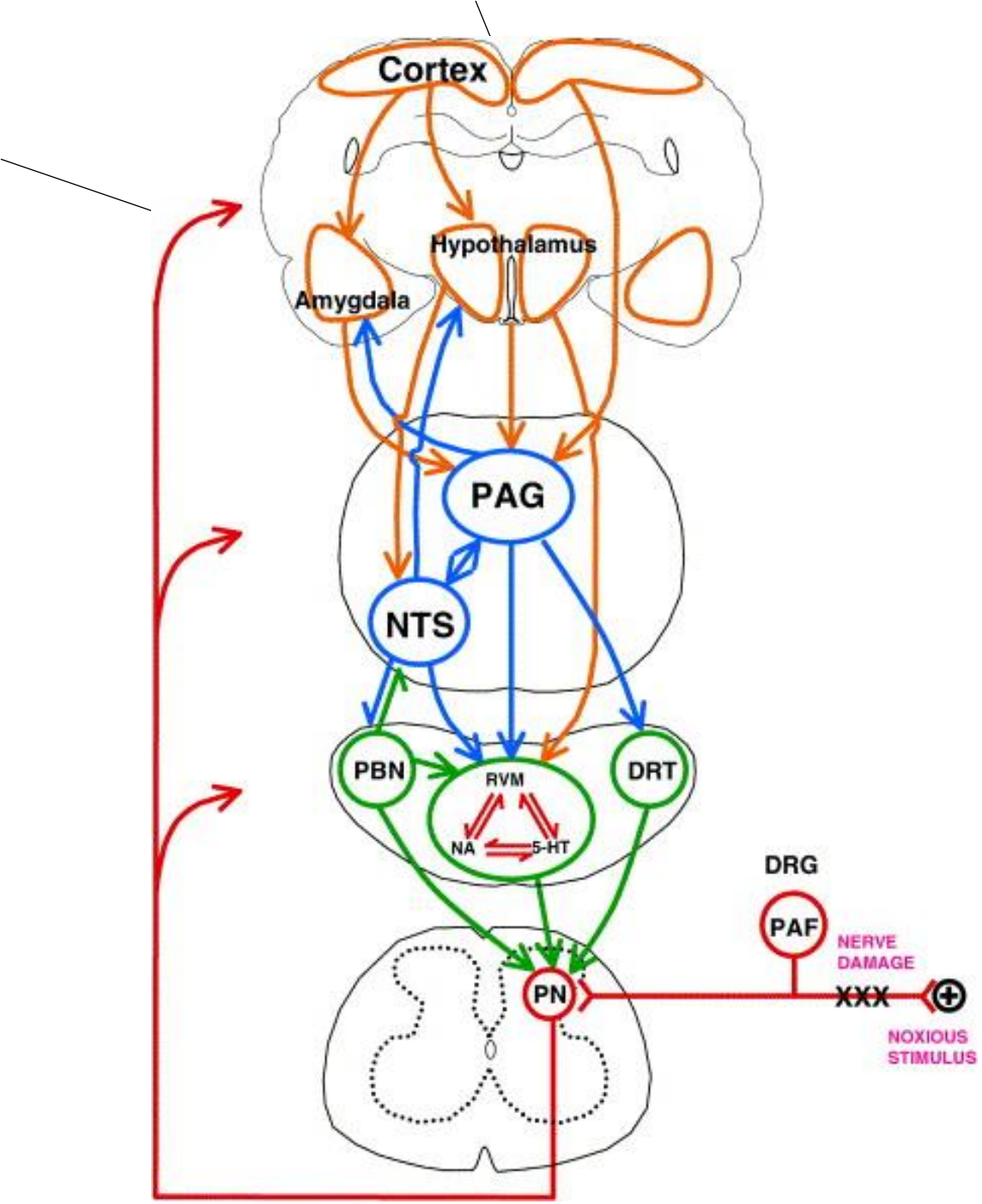
The Chemistry of Descending Control




Cortex
Thalamus
Periaqueductal grey (PAG)
Rostral ventral medulla (RVM)
Dorsal root ganglion (DRG)


B
Dorsal horn
Intermediate zone
Ventral horn


Central perception
Cortex
Tissue injury
Inflammation
Substance P
Prostaglandins
Serotonin
Acetylcholine
Mediator release
Thalamus
Inhibitory mediators: opioid peptides, norepinephrine, glycine, and GABA
Descending inhibitory pathway
Spinothalamic tract
Peripheral stimulus
Nociceptor
Brain stem
Signal transduction
Dorsal root
Dorsal root ganglion
Peripheral nerve
Conduction
Transmission
Spinal cord



Descending Control:

Cerebral structures involved in the initiation and modulation of descending controls of nociceptive information. Note the strategic location of the periaqueductal grey (PAG) and the reciprocal nature of many interconnecting pathways.

CX, cortex

NTS, nucleus tractus solitarius;

PBN, parabrachial nucleus;

DRT, dorsoreticular nucleus;

RVM, rostroventral medulla;

NA, noradrenaline

5-HT, serotonergic

PAF, primary afferent fibre and

DRG, dorsal root ganglion.

PN - projection neurones (PNs)

RACING DRIVER (SAMIR) & CO-DRIVER (VIVEK)


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HiveFunnies

RACING DRIVER (SAMIR) & CO-DRIVER (VIVEK)

- Was Samir the perfect driver / Was Vivek the perfect co-pilot?
- Was the race well controlled?
- Did the car break? (no it didn’t)

What Is Chronic Pain?

Acute Pain: short-term pain that acts as an alarm telling us something is wrong, like breaking your leg.

Chronic Pain: persistent pain that often serves no useful purpose.

Chronic pain is usually defined as: **Pain that persists beyond the normal time that tissues take to heal following an injury.**

Deemed chronic after 3 months.



PAIN AND THE NEUROMATRIX IN THE BRAIN, MELZACK, 2001

The neuromatrix theory of pain

- **multidimensional experience with a “neurosignature” pattern**
  - **may be triggered by sensory inputs**
  - **or generated independently of them!**
- **Chronic pain syndromes, which are often characterized by severe pain associated with little or no discernible injury or pathology, remain a mystery. Pain arises from a widely distributed neural network in the brain rather than directly by sensory input evoked by injury, inflammation, or other pathology.**
- **Patterns involve activating perceptual and behavioural programs after injury, pathology, or chronic stress.**
- **Neuromatrix – genetics modified by sensory experience resulting in a neural pattern that produces pain.**

Melzack, R., 2001. Pain and the neuromatrix in the brain. Journal of dental education, 65(12), pp.1378-1382.

PAIN AS A MULTISYSTEM BODY RESPONSE TO DANGER OR DAMAGE

&

BASIC RESPONSES TO THE EXPERIENCE OF PAIN AND FACTORS THAT LEAD TO THE DEVELOPMENT OF CHRONIC PAIN.

NAIL IN THE HEAD

NAIL IN THE HEAD

- Is it really about the nail?
- What if there was no nail, and she still has symptoms?

- What if someone has a scan showing mild osteoarthritis, but the symptoms are disproportionate to the scan?

There is a high, hard ground where practitioners can make effective use of research-based theory and technique, and there is a swampy lowland where situations are confusing "messes" incapable of technical solution.,...In the swamp are the problems of greatest human concern.

- Donald Schön -

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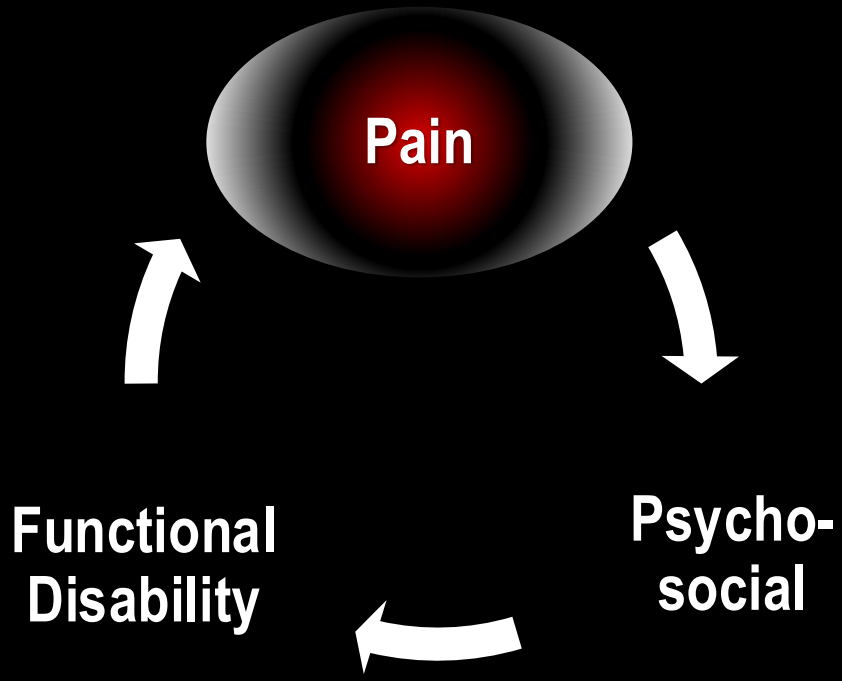
WHY THE THE BIOPSYCHOSOCIAL MODEL (BPS) WORKS

Now we move from *how* the nerves fire to *who* the person is. The BPS model explains why the descending pathways behave the way they do.

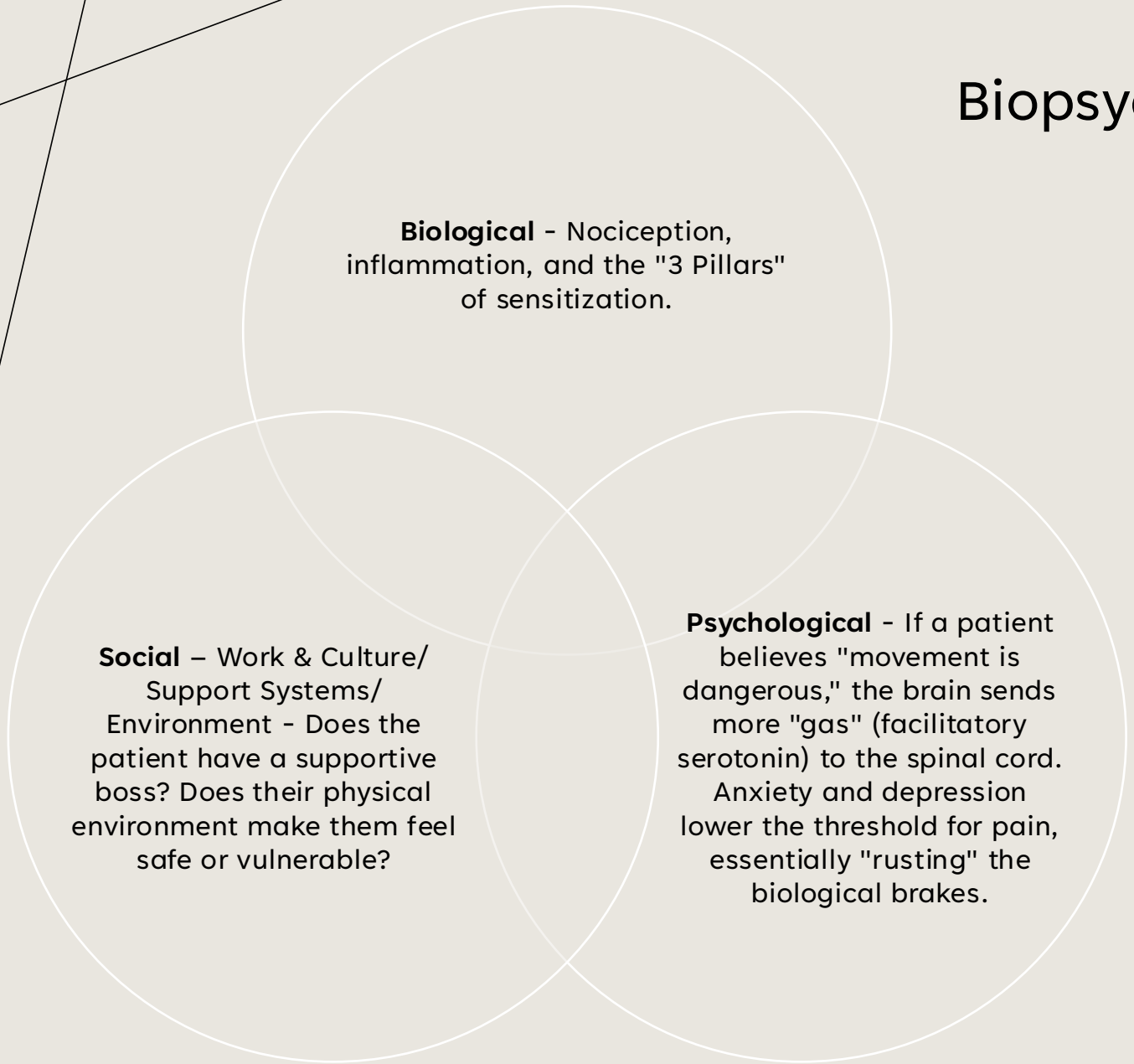
BIOPSYCHOSOCIAL MODEL OF PAIN

PAIN

Not just a symptom
but your patient's lived experience



Biopsychosocial Model (BPS)



Pain is a Lived Experience

Pain is not just a "biological" event. It is the intersection of three distinct domains.

NOCIPLASTIC PAIN

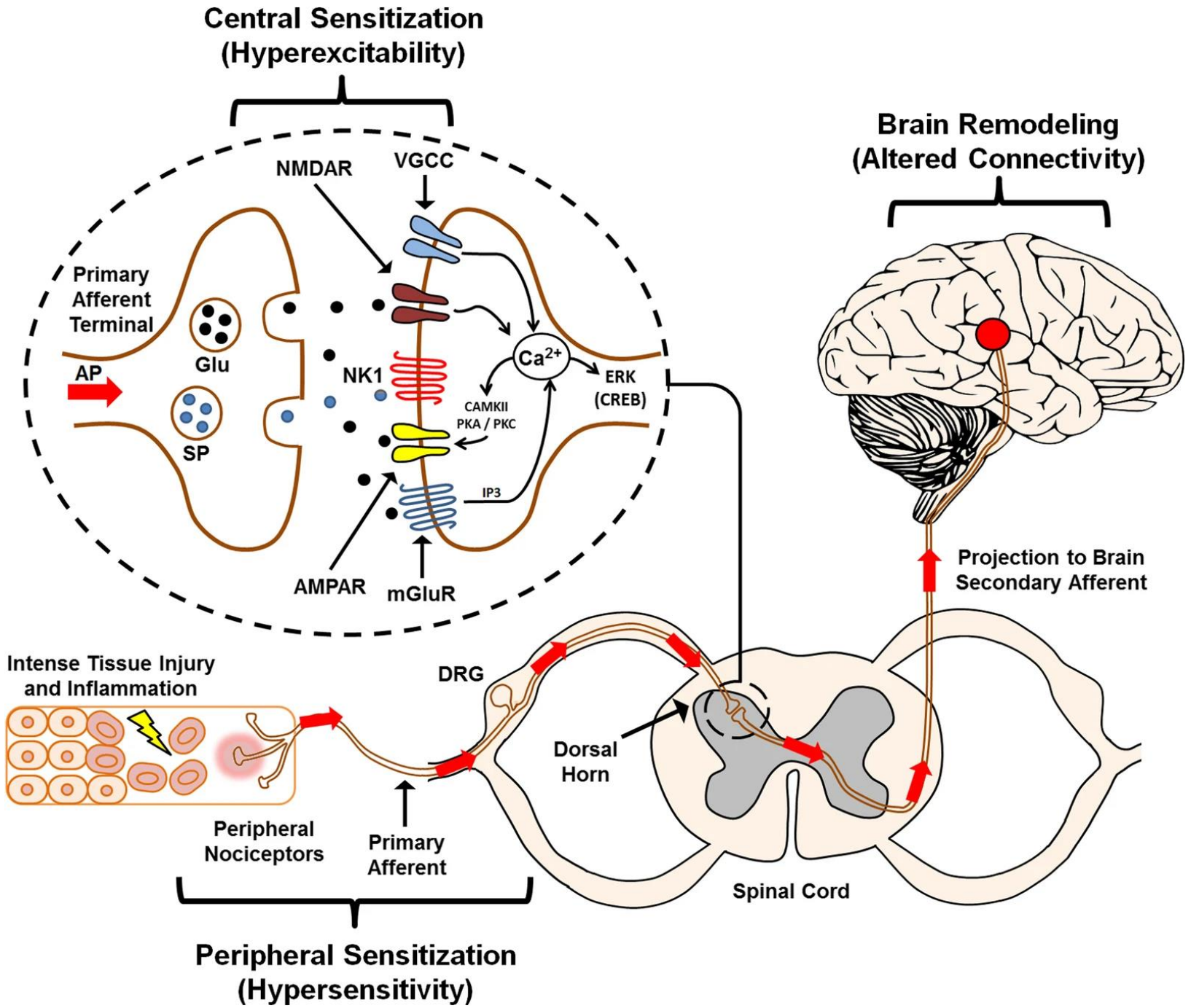
Nociplastic pain as part of a continuum.

- Nociplastic pain as a unique category devoid of actual or threatened tissue or somatosensory damage, there is evidence for the overlap of nociceptive, neuropathic, and nociplastic pain, indicating that nociplastic pain is not a distinct entity, but part of a chronic pain continuum.

- The biopsychosocial model is particularly relevant for nociplastic pain disorders. Although often sought out, a causal factor is usually not evident. Common predisposing factors include a family history of pain, a history of pain experiences, and psychosocial factors such as psychological, emotional, sexual, or physical abuse, or a combination of these.

- We used to think of Nociceptive (tissue), Neuropathic (nerve), and Nociplastic (brain) pain as three separate islands. Most chronic pain patients live on a **continuum**. A patient might start with a nociceptive injury (a disc bulge), develop neuropathic features (sciatica), and eventually transition into a dominant **Nociplastic state** (central sensitization).

The Chemistry of the ‘’Gas’’ Pedal



Pain related to damage of somatic or visceral tissue as a result of trauma or inflammation

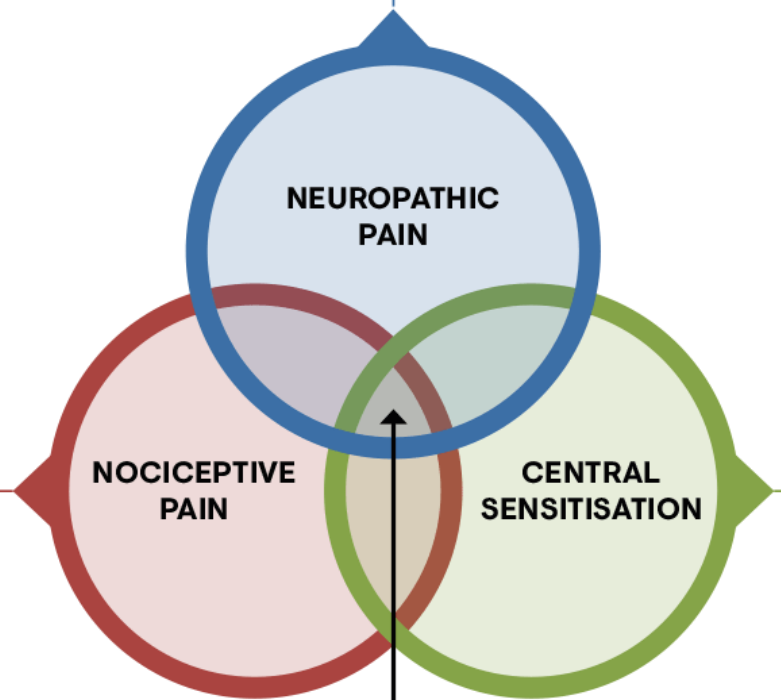
NOCICEPTIVE PAIN

Pain related to damage of peripheral or central nerves

NEUROPATHIC PAIN

Pain without identifiable nerve or tissue damage thought to result from persistent neuronal dysregulation

CENTRAL SENSITISATION



PREDOMINANTLY NEUROPATHIC

- PHN
- pDPN
- Lumbar or cervical radioculopathy
- Stenosis
- Tumour-related neuropathy
- Chemotherapy-induced neuropathy
- Small fibre neuropathy
- Persistent postoperative pain
- Multiple sclerosis pain
- Post-stroke pain
- Pain associated with spin cord injury

PREDOMINANTLY NOCICEPTIVE

- Osteoarthritis
- Rheumatoid arthritis
- Tendonitis, bursitis
- Ankylosing spondylitis
- Gout
- Neck and back pain with structural pathology
- Tumour-related nociceptive pain
- Sickle cell disease
- Inflammatory bowel disease

PREDOMINANTLY CENTRAL SENSITISATION

- Fibromyalgia
- Irritable bowel syndrome
- Tension-type headaches
- Interstitial cystisis/pelvic pain syndrome
- Tempo-manibular joint disorder
- Chronic fatigue syndrome
- Restless leg syndrome
- Neck and back pain without structural pathology

Mixed pain conditions with multiple pain pathophysiologies such as chronic low back pain

PHYSIO TIME

EFFECTS OF MANUAL THERAPY

The theory is that, when we massage or articulate an area, we are activating large myelinated fibers that cause a dampening effect on the neurological C- pain fibres.

This changes the pain threshold of the nervous system and a ‘closing’ of the gating system.

Another theoretical part of the pain gate theory is the descending pathway inhibition via the periaqueductal grey (PAG). This suggests that stimulation of the periaqueductal grey matter of the midbrain causes a response in the descending pathways to the dorsal horn of the spinal cord that stimulates inhibiting interconnecting neurons situated in the substantia gelatinosa.

References: Bialosky, J.E., Beneciuk, J.M., Bishop, M.D., Coronado, R.A., Penza, C.W., Simon, C.B. and George, S.Z., 2018. Unraveling the mechanisms of manual therapy: modeling an approach. journal of orthopaedic & sports physical therapy, 48(1), pp.8-18.

Bialosky, J.E., Bishop, M.D., Price, D.D., Robinson, M.E. and George, S.Z., 2009. The mechanisms of manual therapy in the treatment of musculoskeletal pain: a comprehensive model. Manual therapy, 14(5), pp.531-538.

EFFECTS OF MANUAL THERAPY

Manual therapy is a **sensory input** used to create a **neurological output.** It modulates pain via three primary mechanisms:

1. **The Spinal Level (Gate Control)**
   A. MT activates large-diameter, myelinated A-beta fibers (mechanoreceptors).
   B. These fibers provide excitatory input to inhibitory interneurons in the substantia gelatinosa.
   C. This "closes the gate," preventing nociceptive C-fiber signals from reaching the second-order projection neurons.
2. **The Supraspinal Level (The Top-Down Brakes)**
   A. The rhythmic and novel sensory input of MT is processed by the brain as "safe."
   B. This activates the Periaqueductal Gray (PAG) and the Rostroventral Medulla (RVM).
   C. Triggers the Descending Inhibitory Pathway, releasing endogenous opioids and serotonin to dampen systemic pain sensitivity.
3. **The Peripheral Level (Chemical Washout)**
   A. Physical movement of the tissues and improved circulation.
   B. Reduces the concentration of inflammatory mediators (the "chemical soup") around the nerve endings.
   C. Decreases Peripheral Sensitization, making the "alarm" (nociceptors) less twitchy and harder to trigger.

4. ***BONUS* The Psychological Level (Placebo & Context)**
   A. The therapeutic alliance and the "human touch."
   B. Positive expectations and a reduction in threat perception.
   C. Modulates the Neuromatrix, reducing the brain's overall "danger" evaluation.

EFFECTS OF MANUAL THERAPY

Linking this to manual therapy:

Bialosky, J.E., Beneciuk, J.M., Bishop, M.D., Coronado, R.A., Penza, C.W., Simon, C.B. and George, S.Z., 2018. Unraveling the mechanisms of manual therapy: modeling an approach. journal of orthopaedic & sports physical therapy, 48(1), pp.8-18.

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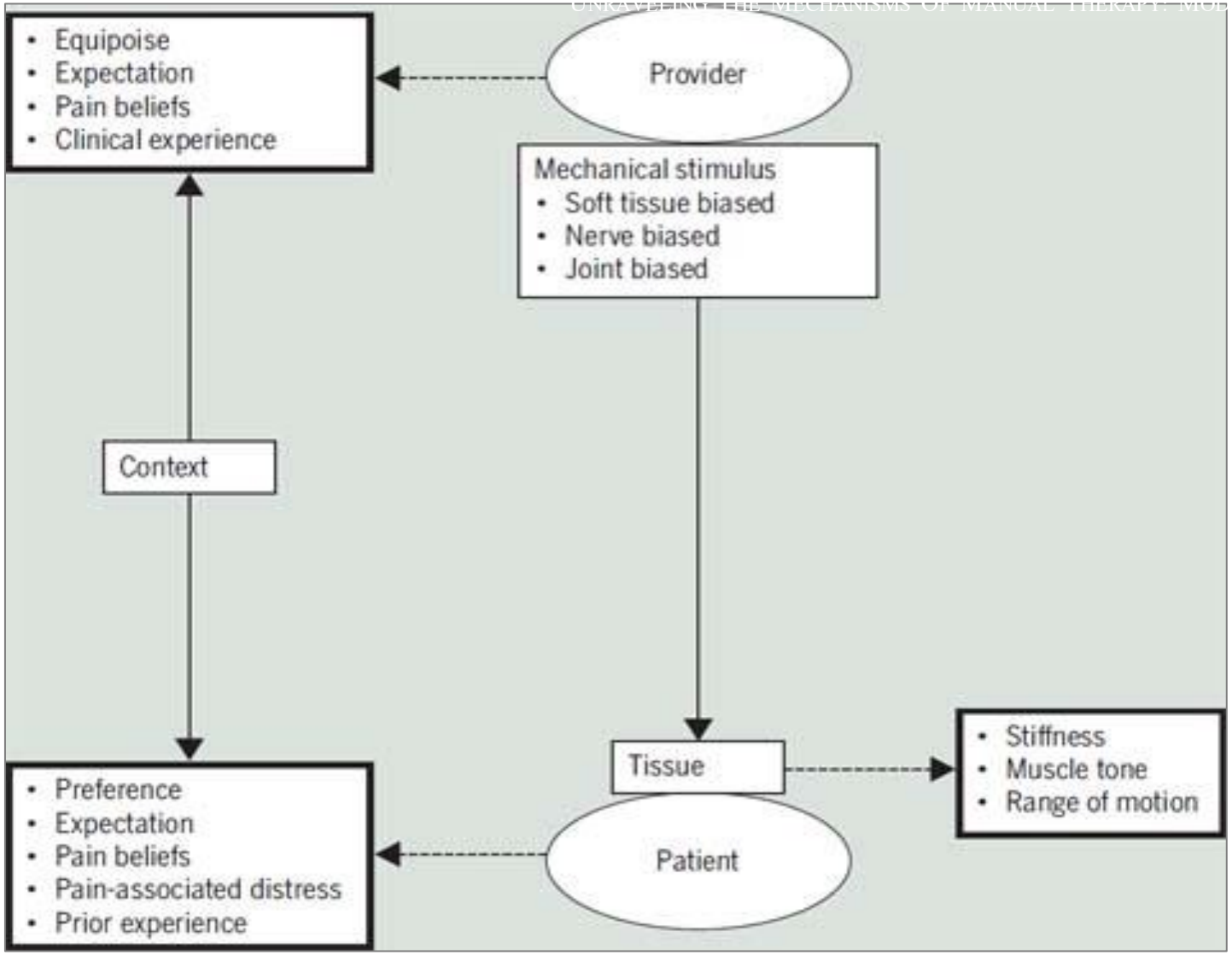


FIGURE 3. Zone 1 of the model, encompassing the interaction between the provider and patient, as well as the mechanical stimulus to the targeted tissue. Solid arrows denote a direct mediating effect. Broken arrows denote an association between a construct and its measure. Bold boxes provide examples of measurable constructs for consideration.

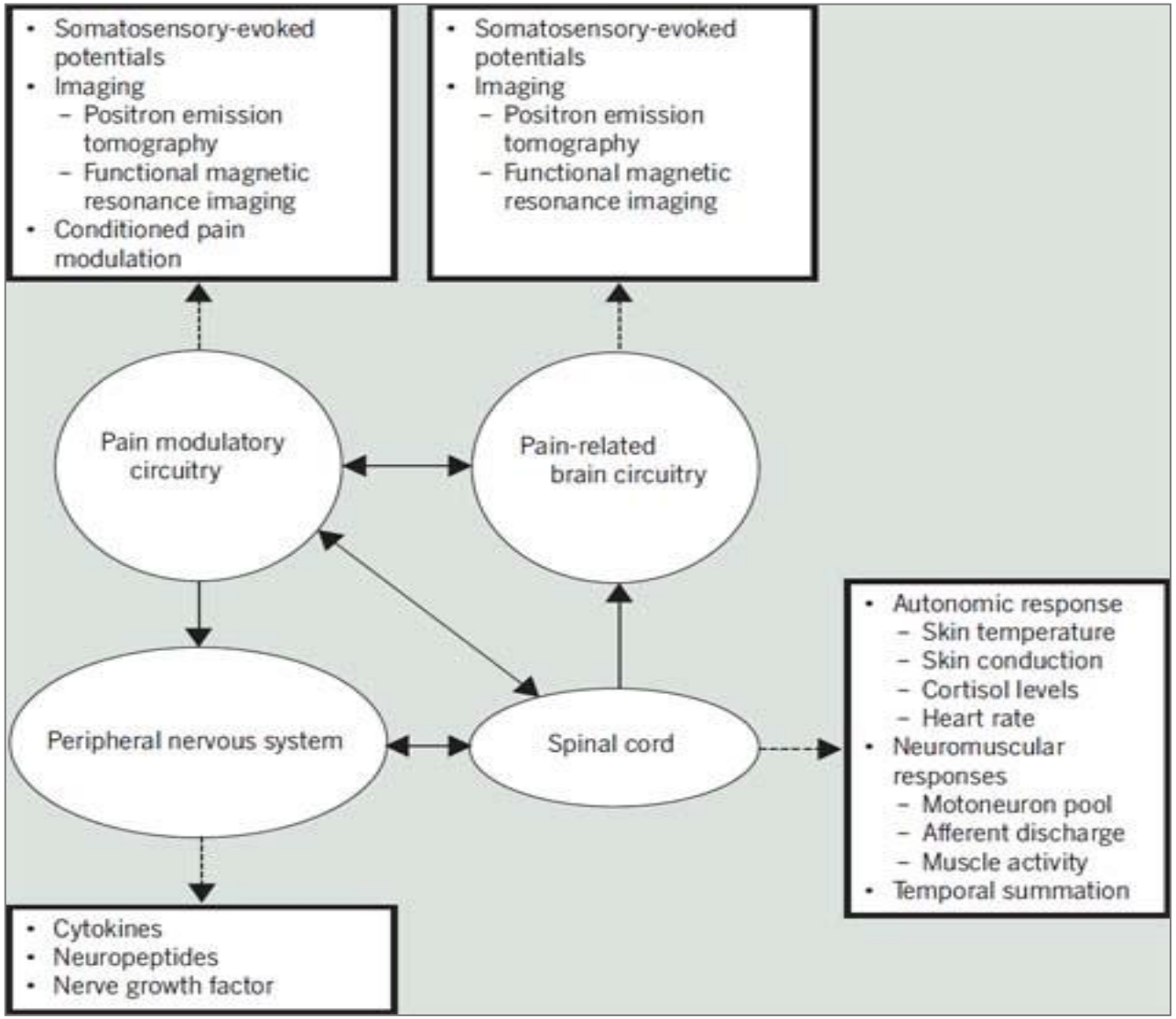


FIGURE 4. Zone 2 of the model, encompassing the nervous system response of the patient to receiving a manual therapy intervention. Solid arrows denote a direct mediating effect. Broken arrows denote an association between a construct and its measure. Bold boxes provide examples of measurable associated responses suggestive of a direct nervous system response.

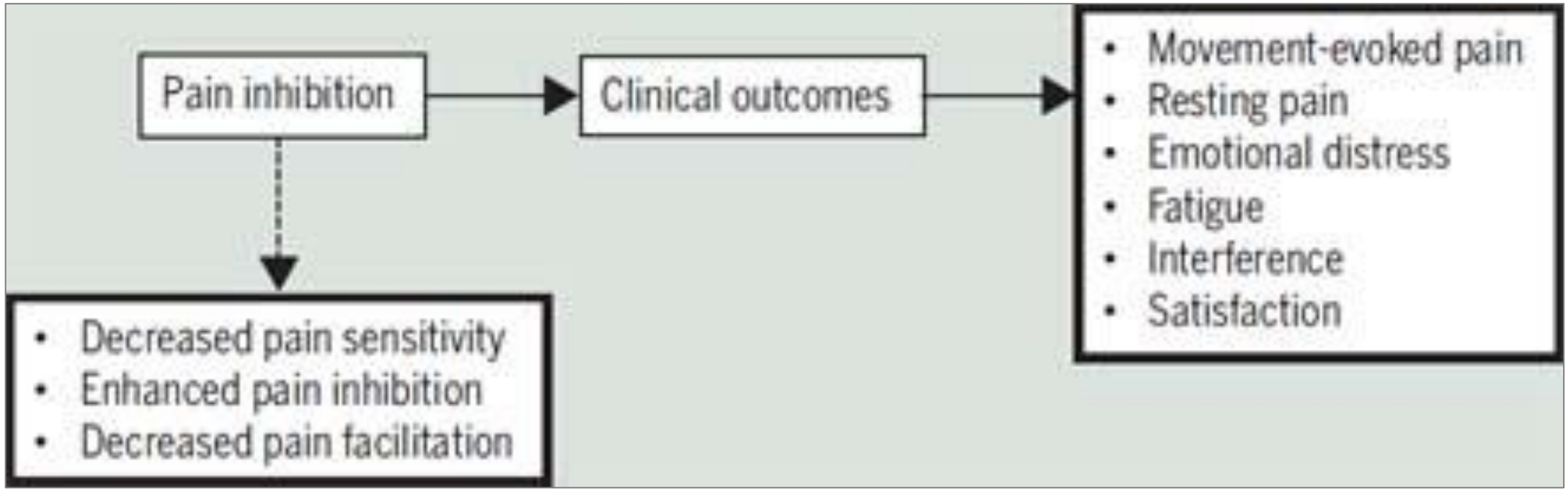
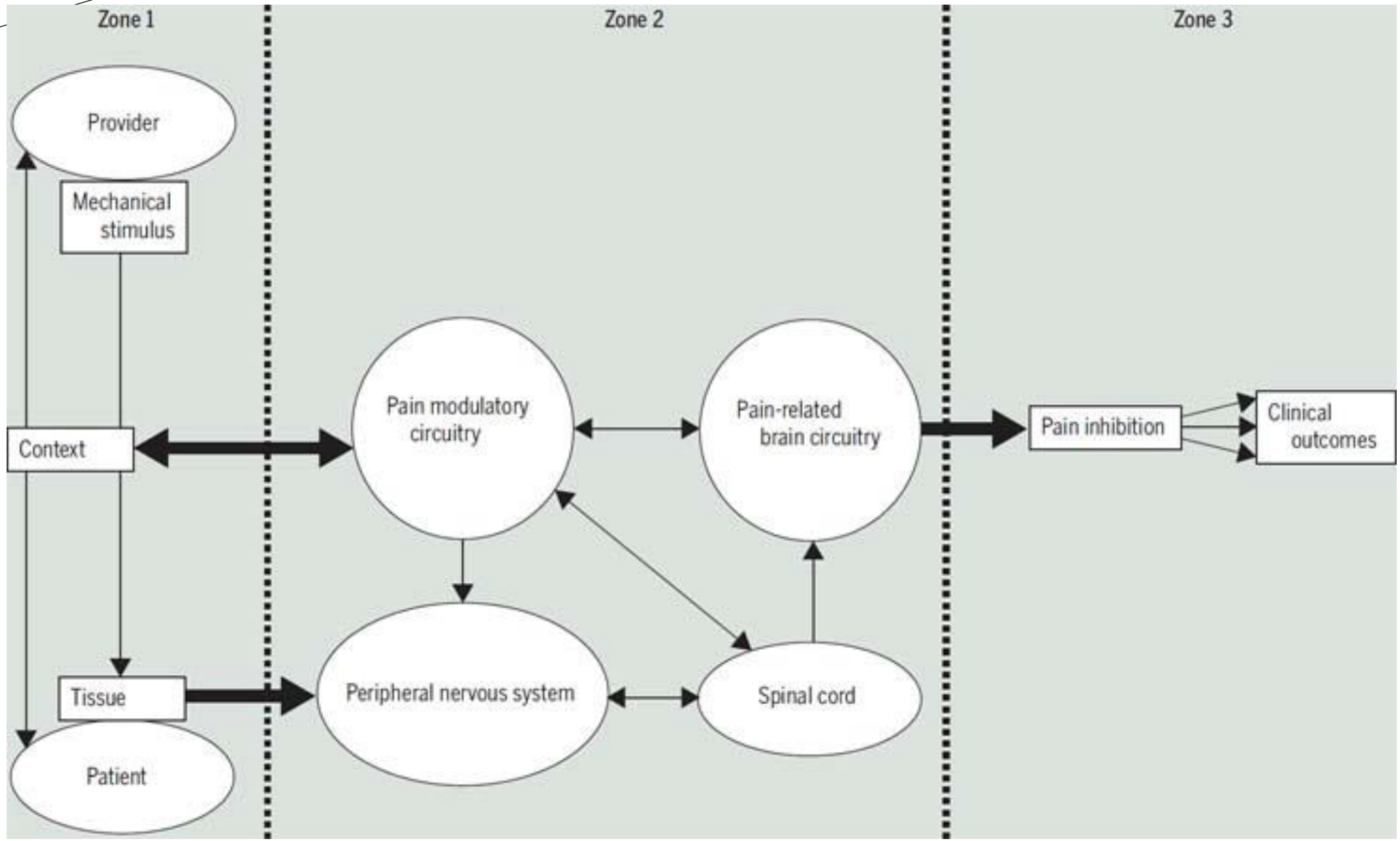
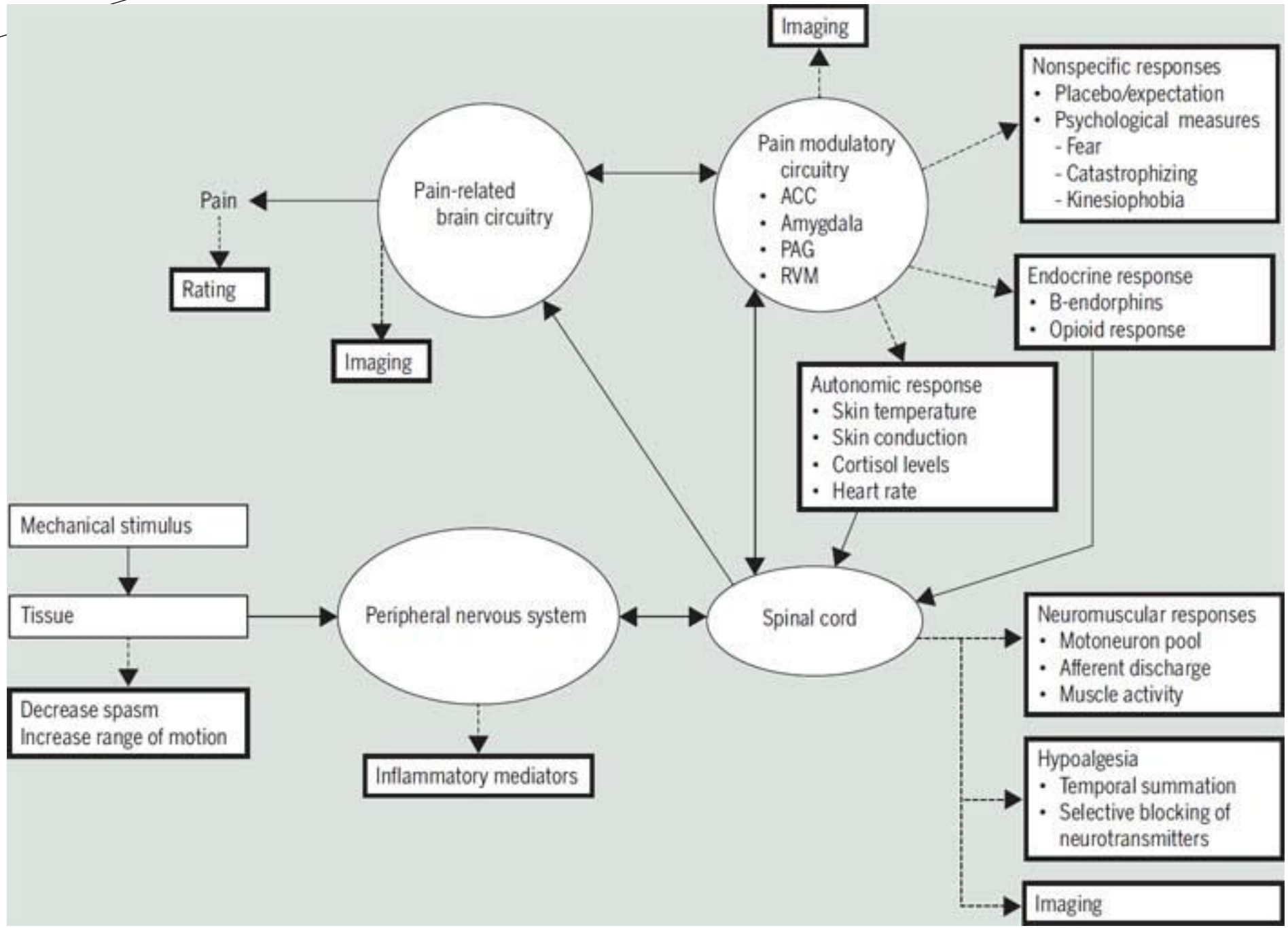


FIGURE 5. Zone 3 of the model, encompassing the outcomes of the model for which related mechanisms of manual therapy may account. Solid arrows denote a direct mediating effect. Broken arrows denote an association between a construct and its measure. Bold boxes provide examples of measurable constructs for consideration.


Zone 1
Zone 2
Zone 3
Provider
Mechanical stimulus
Context
Tissue
Patient
Pain modulatory circuitry
Pain-related brain circuitry
Peripheral nervous system
Spinal cord
Pain inhibition
Clinical outcomes

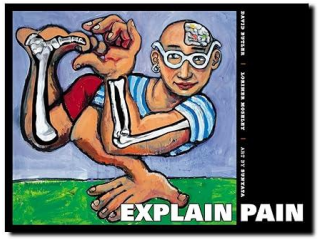

Imaging
Nonspecific responses
• Placebo/expectation
• Psychological measures
- Fear
- Catastrophizing
- Kinesiophobia
Pain
Pain-related brain circuitry
Pain modulatory circuitry
• ACC
• Amygdala
• PAG
• RVM
Rating
Imaging
Endocrine response
• B-endorphins
• Opioid response
Autonomic response
• Skin temperature
• Skin conduction
• Cortisol levels
• Heart rate
Mechanical stimulus
Tissue
Peripheral nervous system
Spinal cord
Neuromuscular responses
• Motoneuron pool
• Afferent discharge
• Muscle activity
Decrease spasm
Increase range of motion
Inflammatory mediators
Hypoalgesia
• Temporal summation
• Selective blocking of neurotransmitters
Imaging

EVIDENCE BASED STRATEGIES FOR CHRONIC PAIN

Since chronic pain is a **multisystem "software" glitch**, our treatment must address the biological, psychological, and social drivers simultaneously.

1. Pain Neuroscience Education
2. Graded Exposure & Exercise
3. Cognitive Functional Therapy (CFT) & CBT

EVIDENCE BASED STRATEGIES FOR CHRONIC PAIN



Since chronic pain is a **multisystem "software" glitch**, our treatment must address the biological, psychological, and social drivers simultaneously.

1. **Pain Neuroscience Education**
    - Hurt does not equal Harm (The Tissue-Brain Gap):
        - Educate patients that pain is a protective output from the brain, not a reliable measure of tissue damage. Just because it feels "worse" doesn't mean the tissues are tearing further. Understanding this reduces the "threat value" and allows the brain to turn down the descending "gas."
    - The Over-Protective Alarm (Sensitization):
        - Explain that in chronic pain, the nervous system has become hyper-vigilant. Much like a home security system that goes off when a leaf touches the window, their nerves have lowered their firing threshold. The goal of therapy is to "re-train" the alarm to be less twitchy, not just to fix a "broken" part.
    - "Motion is Lotion" (Graded Exposure):
        - Help patients move through the "Edge of Discomfort" without fear. If the patient understands that the pain is a "false alarm" from a sensitized system, they can engage in Graded Exposure. This proves to the brain that movement is safe, which eventually triggers neuroplastic changes to "un-smudge" the brain's body maps.
    - The Bio-Plastic Brain (Hope for Change):
        - Emphasize that the nervous system is malleable. Just as it "learned" to be in pain through neuroplasticity (The 3 Pillars), it can "un-learn" pain through new experiences, better sleep, stress management, and safe movement. The system is not "stuck"; it is simply in a state of temporary adaptation.
    - A great reference: Moseley, G. L., & Butler, D. S. (2017). *Explain Pain Supercharged*. Noigroup Publications.

EVIDENCE BASED STRATEGIES FOR CHRONIC PAIN

2. Graded Exposure & Exercise

- Movement is the strongest stimulator of the brain's internal pharmacy (Descending Inhibition).
- Start with movements that feel "safe" to the brain and gradually increase intensity.
- Exercise is the Gold Standard for managing chronic low back pain, as it helps un-smudge the brain's body maps (Cortical Reorganization).

Ideas to begin with patients:

- Yoga flows – together, tons of resources free on Youtube for HEP
- Cycling – low impact, minimal joints moving
- Increase your steps by 1,00 per day – ex: parking car farther from entrance, taking stairs at the tube instead of escalator once per day, 10 minute walk after lunch
- Hydrotherapy - Walking in a pool or gentle water aerobics.
- Movement Snacks – instead fa. 20 min workout, do 2 mins of gentle stretching or chair yoga
- External Focus Games – obstacle course, Balloon volleyball, throwing a beanbag, or using a Wii Sports/VR game

EVIDENCE BASED STRATEGIES FOR CHRONIC PAIN

3. Cognitive Functional Therapy (CFT) & CBT

- To address Pillar 3 (Neuroplasticity) and the Cognitive-Emotional link.
- Modern trials show that targeting *how* a patient thinks about their movement (CFT) is significantly more effective for long-term recovery than just stretching or strengthening the muscles. (O'Sullivan et al., 2018; Williams et al., 2020).

References:
O'Sullivan, P. B., et al. (2018). Cognitive Functional Therapy: An Integrated Behavioral Approach for the Targeted Management of Clinical Back Pain. *Physical Therapy*.
Vibe Fersum, K., et al. (2013). Efficacy of classification-based cognitive functional therapy in patients with non-specific chronic low back pain: A randomized controlled trial. *European Journal of Pain*.
Williams, A. C., et al. (2020). Psychological therapies for the management of chronic pain (excluding headache) in adults. *Cochrane Database of Systematic Reviews*.

EVIDENCE BASED STRATEGIES FOR **FUNCTIONAL NEUROLOGIAL DISORDER (FND)**

Because FND is a disorder of **voluntary control and agency** (the brain is "glitching" the movement), the goal of treatment shifts from "desensitization" to **"retraining the map."**

1. ~~Pain Neuroscience Education~~ → Education on FND
    1. Instead of focusing on why it hurts, you focus on the "Software vs. Hardware" analogy.
    2. The most evidence-based first step for FND is clear communication of the diagnosis. The patient must understand that their nerves are intact (Hardware), but the brain's signaling is blocked or hijacked by a "functional" loop (Software).
2. ~~Graded Exposure & Exercise~~ → Redirecting Attention
    1. In FND, the brain is "over-monitoring" the limb. If an FND patient focuses too hard on moving their leg, the tremor or weakness often gets worse.
    2. We use distraction or external focus of attention.
3. Cognitive Functional Therapy (CFT) & CBT

The current "best" practice for FND (based on the Consensus Recommendations by Nielsen et al., 2015) is **Physiotherapy based on the retraining of movement through distraction.**
We use movements that the patient *can* do (like walking backward or side-stepping) to show the brain it is still capable of movement. This breaks the "learned" pattern of dysfunction.

MAO, J., 2017. CHALLENGES OF MANAGING CHRONIC PAIN. BMJ, 356.

The idea of a comprehensive pain management strategy is now more valuable than ever. Primary care clinicians, surgeons, and emergency department doctors are in the frontline of pain management, and they have a pivotal role in initiating and developing a long term strategy that includes prompt referral to pain specialists. Other key components of a comprehensive management plan include treating underlying conditions; using multiple drugs, including muscle relaxants, antidepressants, anticonvulsants, and topical agents as well as conventional analgesics, to optimise the effectiveness and minimise side effects[8]; adding psychological interventions and physical therapy; and considering integrative approaches such as acupuncture and mind-body therapy.[9] [10]

ADDITIONAL THOUGHTS

Effects of Manual Therapy?

Area of growth for Physiotherapy?

‘Re-modelling’ of brain - postive neuroplastic changes / reversing nociplastic changes